When Curvature Beats Dimension: Euclidean Limits and Hyperbolic Design Rules for Trees

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Abstract

When a tree must be represented by coordinates, how much Euclidean dimension is needed, and what can two hyperbolic coordinates achieve instead? We study this question for finite trees with positive edge lengths. A packing argument gives explicit Euclidean distortion lower bounds from sets of well-separated vertices. For a balanced b-ary tree of height h and edge-weight ratio γ, it yields $\mathrm{dist}(\psi) \geq (b^{\lfloor h/2 \rfloor/d} - 1)/(2\gamma)$ in $\mathbb{R}^d$. A separation-and-radius formulation extends the bound to unbalanced trees without an effective branching approximation. We then give a weighted angular construction in the hyperbolic plane, prove its distortion guarantee, and derive sufficient curvature and radius budgets for a prescribed distortion and absolute separation. Independent edge perturbations admit an exponential confidence bound; explicit dependence assumptions give corresponding extensions. Reproducible experiments evaluate the resulting bounds and embeddings using every pair of vertices.

1 Introduction

A hierarchy can be easy to store as a tree and difficult to represent by a few Euclidean coordinates. Branching creates many vertices that are far apart in the tree but remain within a modest distance of its root. If an embedding preserves those distances, it must fit many separated points into a bounded region. In a fixed Euclidean dimension, that region eventually runs out of room.

This matters when coordinates are used for retrieval, visualization, or a learned representation. Increasing dimension can reduce distortion, but it increases storage and the cost of distance calculations. Hyperbolic space offers another option: its exponential volume growth can accommodate branching in two dimensions. The relevant question is therefore quantitative. Given a tree, a distortion target, and a resource budget, what can be ruled out and what can be constructed?

We answer this question through two complementary bounds. The first is a lower bound on the distortion of *every* Euclidean embedding. The second is an upper bound for a particular hyperbolic embedding. They have different roles. A dimension lower bound can reject an insufficient Euclidean budget. A constructive upper bound can establish that a hyperbolic embedding meets the target. Between these conclusions there can be a gap, which can be addressed by constructing and evaluating candidate embeddings.

The balanced case makes the comparison transparent. A complete b-ary tree of height h contains $b^{\lfloor h/2 \rfloor}$ leaves whose pairwise distances are comparable to its diameter. Packing their images into $\mathbb{R}^d$ forces distortion to grow at least as $b^{\lfloor h/2 \rfloor/d}$, up to the weight ratio and an additive constant. In contrast, the hyperbolic construction below achieves any prescribed distortion greater than one in two dimensions by choosing a sufficiently large distance scale. That scale determines the radius and coordinate precision required by the embedding.

For arbitrary weighted trees, we express the Euclidean bound through the separation and enclosing radius of a selected vertex set. This lets the bound reflect imbalance without replacing the tree by a nominal branching factor. The hyperbolic construction has explicit constants and a guarantee for every pair of vertices. We also analyze bounded random edge weights, with separate guarantees for independent weights, local dependence, and mixing.

Our experiments examine these statements on balanced trees, trees with a long spine and a branching component, and trees with clustered edge weights. We measure worst-case distortion over all vertex pairs, including edges and ancestor–descendant pairs. This connects the empirical quantity directly to the theorems. The main text gives the results and their interpretation; the full proofs and implementation details appear in the appendix.

2 Setting and relation to prior work

Let $T = (V, E, w)$ be a finite rooted tree with at least two vertices and positive edge lengths. Its path metric is d_T. We distinguish the combinatorial height h, which counts edges, from the weighted height

$$H = \max_{v \in V} d_T(o, v),$$

where o is the root. Write $w_{\min} = \min_e w_e$, $w_{\max} = \max_e w_e$, and $\gamma = w_{\max}/w_{\min}$. The complete b-ary tree of height h has exactly b children at every nonleaf vertex and all leaves at depth h.

For an injective map f into a metric space (Y, d_Y), define

$$\operatorname{dist}(f) = \frac{\max_{u \neq v} d_Y(f(u), f(v))/d_T(u, v)}{\min_{u \neq v} d_Y(f(u), f(v))/d_T(u, v)}. \tag{1}$$

A noninjective map has infinite distortion. Distortion is unchanged by multiplying every target distance by the same positive constant. For Euclidean lower bounds, we rescale the map so that its largest distance ratio is exactly one. A distortion-C map then satisfies

$$\frac{d_T(u, v)}{C} \leq \|f(u) - f(v)\| \leq d_T(u, v).$$

All results concern the entire vertex set unless a subset is specified explicitly.

Euclidean metric embedding theory provides general distortion bounds and methods for relating graph structure to embedding difficulty [1–3]. Our Euclidean argument is a direct application of volume packing.

Sarkar [4] established low-distortion tree embeddings in the hyperbolic plane, and Sala et al. [8] studied the tradeoff between dimension and precision. Our angular construction follows the same geometric principle, with explicit constants for weighted trees. These constants connect the distortion guarantee to curvature, radius, and absolute separation requirements.

Learned hyperbolic representations address related questions through optimization, including Poincaré embeddings, entailment cones, and hyperbolic graph convolution [5–7]. Those methods may optimize task-specific losses rather than worst-case metric distortion. Here the tree metric is given, so the construction and its pairwise guarantee can be evaluated directly.

3 Euclidean limits from separation and radius

3.1 The balanced-tree bound

We start with the case in which branching and depth have a direct interpretation.

Corollary 3.1 (Balanced trees)**.** *Let $b \geq 2$ and $h \geq 2$. Every embedding of a complete weighted b-ary tree of height h into $\mathbb{R}^d$ has distortion*

$$C \geq \max\left\{1, \frac{b^{\lfloor h/2 \rfloor/d} - 1}{2\gamma}\right\}. \tag{2}$$

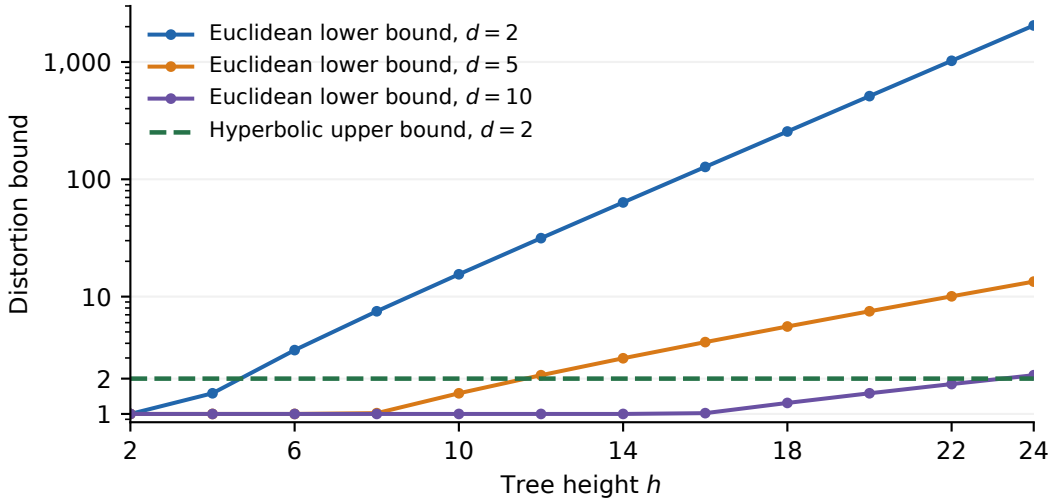


Figure 1: Analytical bounds for complete 4-ary trees with unit edge lengths. Solid curves are the Euclidean lower bound in Eq. (2), including $C \geq 1$. The dashed line is the distortion upper bound of Theorem 4.1 with $\eta = 1/3$. Its two-dimensional construction uses a normalized radius proportional to h. The curves compare a Euclidean lower bound with a hyperbolic construction upper bound.

At fixed dimension and bounded weight ratio, the lower bound grows exponentially with depth. Increasing dimension slows that growth. The factor $1/\gamma$ comes from comparing the shortest guaranteed separation with an upper bound on the enclosing radius.

The argument is geometric. Choose one leaf below each vertex at depth $\lfloor h/2 \rfloor$. These leaves are separated by long paths in the tree. A nonexpansive embedding places them in a Euclidean ball of radius at most $hw_{\max}$, while distortion bounds how close their images can be. Comparing the volumes of small disjoint balls around those images yields Eq. (2). Appendix A gives the complete proof.

For example, take $b = 4$, $h = 12$, and uniform weights. In two dimensions Eq. (2) gives $C \geq 31.5$. A target distortion of five is therefore impossible in $\mathbb{R}^2$. Inverting the same inequality gives the necessary condition

$$d \geq \left\lceil \frac{\lfloor h/2 \rfloor \log b}{\log(1 + 2\gamma C_0)} \right\rceil \tag{3}$$

for a target $C_0 \geq 1$. In this example the right-hand side is four. It excludes dimensions below four; an embedding is still needed to establish what happens at or above four.

3.2 Arbitrary weighted trees

An unbalanced tree need not have a useful nominal branching factor. Instead, we can identify a set of vertices that are separated in the metric itself. For $S \subseteq V$ with $N = |S| \geq 2$, let

$$s(S) = \min_{u \neq v \in S} d_T(u, v), \qquad R(S) = \min_{z \in V} \max_{u \in S} d_T(z, u). \tag{4}$$

Both quantities use the observed edge weights. Restricting centers to vertices makes $R(S)$ straightforward to evaluate; any upper bound on this radius can also be used.

Theorem 3.2 (Packing bound)**.** *For every $S \subseteq V$ with $|S| \geq 2$, a distortion-C embedding of T into $\mathbb{R}^d$ satisfies*

$$C \geq \frac{s(S)}{2R(S)} \left(|S|^{1/d} - 1 \right). \tag{5}$$

The bound depends on how many separated points fit within a given tree radius. A long unary stem does not weaken a witness inside a branching subtree, because its enclosing center can be placed in that subtree. Conversely, many leaves provide little information if their mutual distances are small compared with their enclosing radius.

Weighted binary trees also show that the dependence on edge weights can be attained. For the family in Proposition A.4, which has geometrically decreasing edge lengths, the optimal distortion into the line lies between $N/(6\gamma)$ and $8N/\gamma$, where N is the number of leaves. The packing witness here contains all leaves; the midpoint witness used in Eq. (2) can give a weaker bound.

It is unnecessary to find the best possible set S to obtain a valid bound. Any collection $\mathcal{W}$ of sets with at least two vertices gives the necessary dimension threshold

$$d_{\mathrm{LB}}(T, C_0; \mathcal{W}) = \max\left\{1, \left\lceil \max_{S \in \mathcal{W}} \frac{\log |S|}{\log(1 + 2R(S)C_0/s(S))} \right\rceil\right\}. \tag{6}$$

The strongest available restriction is the maximum over witnesses. Our experiments select leaves below same-depth antichains and compare the resulting weighted branch lengths. Appendix D specifies the finite collection used.

The same packing theorem holds for any finite metric. A tree found inside a graph can supply a witness when graph distances on the selected vertices have a known lower bound relative to their tree distances. A graph minor alone provides no such guarantee, since additional paths can shorten distances. Appendix A states the precise transfer result.

4 A hyperbolic construction with explicit budgets

Hyperbolic space can keep distortion bounded as the tree grows, but the embedding must use a suitable scale. We state that scale explicitly. Let $B \geq 2$ be an upper bound on the number of children of any vertex, and let $\mathbb{H}^2_{-\kappa}$ have curvature $-\kappa$, where $\kappa > 0$.

The construction assigns each vertex an angular interval. A child receives a shorter interval inside its parent's interval, with a gap between neighboring children. The intervals shrink according to weighted root distance. Vertex v is placed at the center angle of its interval and at hyperbolic radius $\tau d_T(o, v)/\sqrt{\kappa}$. The gaps control the distance between different branches; the radial coordinate controls ancestor–descendant distances. Appendix B gives the algorithm and proves the following result.

Theorem 4.1 (Weighted hyperbolic embedding)**.** *Fix $0 < \eta < 1$ and choose*

$$\tau \geq \frac{2\log(4B)}{\eta w_{\min}}. \tag{7}$$

The angular construction gives an embedding $\phi : V \to \mathbb{H}^2_{-\kappa}$ such that, for every $u, v \in V$,

$$(1-\eta)\frac{\tau}{\sqrt{\kappa}} d_T(u,v) \leq d_{-\kappa}(\phi(u), \phi(v)) \leq (1+\eta)\frac{\tau}{\sqrt{\kappa}} d_T(u,v). \tag{8}$$

Its distortion is at most $(1+\eta)/(1-\eta)$, and its maximum root radius is $\tau H/\sqrt{\kappa}$.

To attain a target $C_0 > 1$, set $\eta = (C_0 - 1)/(C_0 + 1)$ and choose τ by Eq. (7). The distortion guarantee does not depend on tree height. Height instead enters the radius through H. The construction applies to heterogeneous weights and unbalanced trees with the same statement.

4.1 Distortion, radius, and absolute separation

Curvature changes the unit of hyperbolic distance: $d_{-\kappa} = d_{-1}/\sqrt{\kappa}$. It therefore changes the physical radius without changing distortion. A radius requirement becomes informative when paired with an absolute distance scale. Suppose that every pair of distinct vertices should be at least $s_0 > 0$ apart and the embedding must lie within radius R. Since $d_T(u, v) \geq w_{\min}$ for distinct vertices, Theorem 4.1 gives the sufficient conditions

$$\frac{\tau H}{R} \leq \sqrt{\kappa} \leq \frac{(1-\eta)\tau w_{\min}}{s_0}. \tag{9}$$

This interval is nonempty whenever

$$R \geq \frac{H s_0}{(1-\eta) w_{\min}}. \tag{10}$$

Equation (10) is a sufficient budget for this construction and guarantee. For a selected pair set whose tree distances are all at least $a > 0$, a can replace $w_{\min}$ in these two conditions.

Numerical scale is a separate consideration. The largest normalized radius is $M = \sqrt{\kappa} R_{\text{used}} = \tau H$. In the unit Poincaré disk, the largest norm is $\tanh(M/2)$. Changing curvature leaves this norm unchanged. Thus a smaller physical radius obtained through larger curvature does not by itself resolve finite-precision problems. Our implementation computes distances in the logarithmic domain and avoids subtracting nearly identical global angles. The numerical procedure and its cost are given in Appendix D.

5 Random edge weights

The deterministic bound applies to every realized set of positive weights. A probabilistic bound can be useful before those weights are observed. Assume $0 < \varepsilon < 1$ and independent edge weights $W_e \in [1-\varepsilon, 1+\varepsilon]$ with $\mathbb{E}W_e = 1$.

Choose N leaves from the topology in advance. Each has a terminal path of r edges, and the path between any two selected leaves must contain both terminal paths. Equal-depth leaves with disjoint terminal paths satisfy this condition. For unequal leaf depths, the condition must be checked directly. Let h be the tree's combinatorial height.

Theorem 5.1 (Independent perturbations)**.** *Fix $0 < \delta \le \varepsilon$ and $0 < \theta < 1$, and put*

$$p = 1 - \exp\left(-\frac{\delta^2 r}{2\varepsilon^2}\right), \qquad m = (1-\theta)pN.$$

With probability at least $1 - \exp(-\theta^2 pN/2)$, every embedding into $\mathbb{R}^d$ has distortion

$$C \ge \frac{(1-\delta)r}{(1+\varepsilon)h}\left(m^{1/d} - 1\right). \tag{11}$$

The event ensures that many terminal paths retain length at least $(1-\delta)r$. Those leaves remain separated and can be used in the packing argument. The theorem does not require the edge distributions to be identical. With $\delta = \varepsilon/2$, the individual bad-path probability is at most $e^{-r/8}$.

Appendix C gives complete proofs and two dependence extensions. A dependency graph leads to explicit penalties for dependence within a terminal path and between terminal paths. Geometric mixing gives a bound through path variances and cross-branch covariances. Their confidence levels are stated separately. When several scales or parameter choices are compared, a finite union bound provides a simultaneous guarantee.

6 Using the bounds to choose a representation

The bounds give a three-step procedure for a target distortion $C_0 > 1$, a Euclidean dimension budget d, and optional hyperbolic radius and separation requirements:

1. Compute Eq. (6) from a collection of separated sets. If $d < d_{\text{LB}}$, no Euclidean embedding in that dimension meets C_0.
2. Choose η and τ for the hyperbolic construction. Check Eq. (9) if a radius and absolute margin are specified, and check the available numerical precision against τH.
3. For any fitted Euclidean or hyperbolic candidate, evaluate Eq. (1). A candidate with distortion at most C_0 certifies that its budget suffices.

If a Euclidean budget passes the lower-bound check but no candidate meets the target, the outcome is unresolved.

Storage and query costs should be compared after these checks. A Euclidean representation uses nd scalar coordinates and a distance evaluation takes $O(d)$ arithmetic operations. A hyperbolic representation uses two mathematical coordinates per vertex, but their required precision can grow with τH. Our tree-aware distance evaluation also uses the tree structure. We therefore report its actual implementation separately from the dimensional comparison.

7 Experiments

We evaluate three questions: whether the construction meets its distortion guarantee on every vertex pair, how dimension and tree shape affect the fitted Euclidean embeddings, and how many separated paths survive independent edge noise.

7.1 Distortion across tree families

Each tree has 127 vertices and 64 leaves. We generate 30 independently weighted instances of each family: a balanced binary tree, a spine with attached binary bushes, and a balanced tree with sharply different weights in its two main branches. Euclidean metric MDS uses dimensions 2, 5, and 10. The hyperbolic construction uses $\eta = 1/3$, which guarantees distortion at most two. Every measurement includes all 8,001 unordered vertex pairs. Appendix D gives the generators, solver settings, and numerical evaluation.

Table 1 reports the results. The hyperbolic construction has distortion between 1.10 and 1.13 across all 90 trees. MDS improves as dimension increases, with the largest remaining distortion on the clustered-weight family. These are the distortions attained by a fixed stress-minimization procedure; the Euclidean lower bounds provide the separate test for impossibility. Parent retrieval is easier than preserving all distances: ten-dimensional MDS has mean Hits@1 of at least 0.998 in every family despite distortion above one (Table 3).

Table 1: All-pairs distortion on 127-vertex weighted trees. Entries are medians over 30 instances, with interquartile ranges below. MDS minimizes raw stress using three starts and at most 1,000 iterations per start.

Embedding	Balanced	Spine-and-bush	Clustered weights
MDS, $d = 2$	37.02 [33.36, 41.91]	43.77 [40.48, 48.39]	38.39 [33.83, 48.05]
MDS, $d = 5$	6.40 [6.11, 6.54]	5.95 [5.72, 6.17]	11.91 [11.10, 12.85]
MDS, $d = 10$	2.96 [2.90, 3.01]	3.10 [3.06, 3.15]	4.33 [4.11, 4.43]
Constructive $\mathbb{H}^2$	1.117 [1.114, 1.121]	1.119 [1.115, 1.122]	1.113 [1.110, 1.115]

7.2 Depth and weight sensitivity

On unit-weight binary trees of heights three through seven, the construction's distortion stays at 1.127, while the distortion of 10-dimensional MDS increases from 1.52 to 3.60. A separate sweep increases the weights in one root branch of a fixed weighted tree. Figure 2 shows both sweeps. The hyperbolic scale is chosen from the smallest edge weight, so increasing heavy-branch weights increases the required normalized radius even when distortion changes little.

7.3 Survival of terminal paths under random weights

We sample 10,000 independent trials at each of nine combinations of depth and noise level. For binary trees with $h \in \{8, 12, 16\}$, we fix $N = 2^{h/2}$ disjoint terminal paths of length $r = h/2$ and draw edge weights uniformly from $[1-\varepsilon, 1+\varepsilon]$, with $\varepsilon \in \{0.1, 0.3, 0.5\}$. A path is good if its weight is at least $(1 - \varepsilon/2)r$. At $\varepsilon = 0.3$, the mean numbers of good paths are 15.32 of 16, 62.98 of 64, and 254.42 of 256. With count slack $\theta = 1/2$, the required counts are 4, 17, and 81. Theorem 5.1 guarantees count-event probabilities of 0.545, 0.985, and more than 0.99999999, respectively. Every sampled trial satisfies its count event.

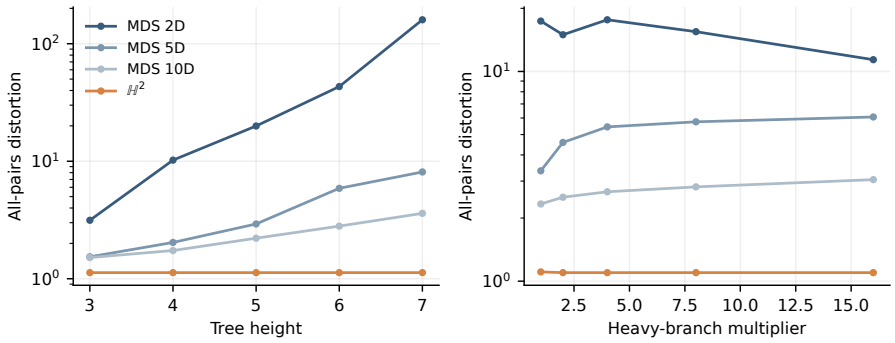


Figure 2: All-pairs distortion under changes in depth and weights. Left: unit-weight binary trees with 15–255 vertices. Right: a fixed 63-vertex binary tree with independently sampled weights in $[1, 2]$; one root branch is multiplied by the indicated factor. Each point is one tree and a three-start MDS fit.

8 Discussion

The comparison is governed by two different geometric quantities. Euclidean packing depends on the number of separated vertices relative to their enclosing radius. Hyperbolic distortion can remain bounded in two dimensions, while the required distance scale and coordinate precision grow with weighted depth and the target accuracy. Keeping these quantities separate gives a useful procedure: reject Euclidean budgets that violate a lower bound, construct embeddings with explicit upper bounds, and measure the remaining cases directly.

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A Euclidean packing and tree witnesses

Throughout this section, $T = (V, E)$ is a finite rooted tree with at least two vertices and positive edge lengths w_e. Its vertex metric is d_T, its combinatorial height is h, and its weighted height is

$$H = \max_{v \in V} d_T(\mathrm{root}, v).$$

All embeddings in the tree results below are defined on every vertex unless an observation set is specified. An embedding of finite distortion is injective. After rescaling, an embedding f of distortion at most C satisfies

$$\frac{d_T(u,v)}{C} \le \|f(u) - f(v)\|_2 \le d_T(u,v) \qquad (u, v \in V). \tag{12}$$

A.1 The packing bound

For a set $S \subseteq V$ with $|S| \ge 2$, define its separation and its enclosing radius by

$$s(S) = \min_{\substack{u,v \in S \\ u \ne v}} d_T(u,v), \qquad R_S = \min_{z \in V} \max_{u \in S} d_T(z,u). \tag{13}$$

These quantities are positive, and the minimum defining R_S is attained because V is finite.

Theorem A.1 (Euclidean packing bound)**.** *Let $S \subseteq V$ contain $N \ge 2$ vertices, and let $0 < s \le s(S)$. Every embedding $f : V \to \mathbb{R}^d$ of distortion C satisfies*

$$N \le \left(1 + \frac{2R_S C}{s}\right)^d, \qquad C \ge \frac{s}{2R_S}\left(N^{1/d} - 1\right). \tag{14}$$

More generally, if $z \in V$ and $d_T(z,u) \le R$ for all $u \in S$, the same statements hold with R in place of R_S.

Proof. Normalize f as in (12). Choose a vertex z attaining R_S and translate the image so that $f(z) = 0$. Then $\|f(u)\|_2 \le R_S$ for every $u \in S$, while distinct selected images have distance at least s/C. The open Euclidean balls of radius $s/(2C)$ about these N images are pairwise disjoint. They are all contained in the ball of radius $R_S + s/(2C)$ about the origin. Comparing their d-dimensional volumes gives

$$N\left(\frac{s}{2C}\right)^d \le \left(R_S + \frac{s}{2C}\right)^d.$$

Division and rearrangement yield (14). The same proof applies to any specified center and valid enclosing radius R. □

Taking $s = s(S)$ proves Theorem 3.2. The proof uses only a metric and a center in the domain of the embedding. In particular, let $O \subseteq V$ be an observed vertex set and let $f : O \to \mathbb{R}^d$. For $S \subseteq O$ with $|S| \ge 2$, define

$$R_{S;O} = \min_{z \in O} \max_{u \in S} d_T(z,u). \tag{15}$$

Then (14) holds with $R_{S;O}$. This version uses the restriction of the tree metric to O and requires no embedding of the unobserved vertices. For example, any selected observed vertex supplies an enclosing radius at most $\mathrm{diam}_T(S)$. A root-based radius H is available whenever the root is included in the embedding domain.

A.2 Antichain witnesses for unbalanced trees

Lemma A.2 (Branch witnesses)**.** *Fix $z \in V$. Let A be an antichain of $N \ge 2$ proper descendants of z: no element of A is an ancestor of another. For each $a \in A$, choose a descendant leaf u_a, and write $p(a)$ for the parent of a. Set*

$$\ell = \min_{a \in A} d_T(p(a), u_a), \qquad R = \max_{a \in A} d_T(z, u_a).$$

Every embedding $f : V \to \mathbb{R}^d$ of distortion C satisfies

$$C \ge \frac{\ell}{R}\left(N^{1/d} - 1\right). \tag{16}$$

Proof. For distinct $a, b \in A$, let c be the lowest common ancestor of u_a, u_b. Incomparability of a and b implies that c is a strict ancestor of both. Therefore c is an ancestor of $p(a)$ and of $p(b)$, and the path from u_a to u_b contains both branch segments $p(a) \rightsquigarrow u_a$ and $p(b) \rightsquigarrow u_b$. These segments have disjoint edge sets, although their initial vertices may coincide. Hence

$$d_T(u_a, u_b) \geq d_T(p(a), u_a) + d_T(p(b), u_b) \geq 2\ell.$$

The N selected leaves lie within distance R of z. Apply Theorem A.1 with separation $s = 2\ell$. □

The lemma uses weighted branch lengths directly and permits leaves at different combinatorial depths. A width profile can be formed by recording the attainable witness sizes N together with their pairs (R, ℓ); both scales enter the resulting lower bound.

A.3 Balanced trees

Corollary A.3 (Balanced-tree lower bounds)**.** *Suppose T is a complete b-ary tree of height $h \geq 2$, where $b \geq 2$, and its edge lengths lie in $[w_{\min}, w_{\max}]$. Put $\gamma = w_{\max}/w_{\min}$. For every $1 \leq k \leq h$ and every embedding $f : V \to \mathbb{R}^d$ of distortion C,*

$$C \geq \frac{(h-k+1)w_{\min}}{H}\left(b^{k/d} - 1\right) \geq \frac{h-k+1}{h\gamma}\left(b^{k/d} - 1\right). \tag{17}$$

In particular,

$$C \geq \frac{1}{2\gamma}\left(b^{\lfloor h/2 \rfloor/d} - 1\right). \tag{18}$$

Proof. The b^k vertices at depth k form an antichain. Select one depth-h leaf below each of them. The path from the parent of a depth-k vertex to its selected leaf has $h - k + 1$ edges, so its weighted length is at least $(h - k + 1)w_{\min}$. All selected leaves have root distance at most $H \leq hw_{\max}$. Lemma A.2 proves (17). For $k = \lfloor h/2 \rfloor$, the factor $(h - k + 1)/h$ is at least $1/2$, giving (18). Together with $C \geq 1$, this proves Corollary 3.1. □

A.4 Weighted hierarchies

Proposition A.4 (Sharpness of radius-based packing)**.** *Let $T_{h,q}$ be a complete binary tree of height $h \geq 1$, where each edge entering depth k has length q^{h-k} and $3/2 \leq q \leq 7/4$. Write $N = 2^h$ and $\gamma = q^{h-1}$. If $C_{\mathbb{R}}(T_{h,q})$ is the infimum of the distortions of embeddings of its vertices into the line, then*

$$\frac{N}{6\gamma} \leq C_{\mathbb{R}}(T_{h,q}) \leq \frac{8N}{\gamma}. \tag{19}$$

Thus the radius-based packing bound using all leaves is attained up to universal constants on this family.

Proof. All leaves have root distance

$$H = \sum_{j=0}^{h-1} q^j = \frac{q^h - 1}{q - 1} \leq \frac{q}{q-1}\gamma \leq 3\gamma.$$

Two leaves in opposite root branches have distance $2H$, so every enclosing tree ball for all leaves has radius at least H. The root supplies such a ball, and hence their enclosing radius is $R_S = H$. Their minimum pairwise distance is 2, because all terminal edges have weight 1 and sibling leaves attain distance 2. Theorem A.1 with $d = 1$ therefore gives

$$C_{\mathbb{R}}(T_{h,q}) \geq \frac{N-1}{H} \geq \frac{N}{6\gamma},$$

using $N \geq 2$.

For the upper bound, perform a depth-first traversal, always visiting the first child before the second child and traversing edges at unit speed with respect to their lengths. Map each vertex v to its first-arrival time $t(v) \in \mathbb{R}$. The arrival times are distinct. For $t(u) < t(v)$, the intervening traversal is a walk from u to v, so

$$t(v) - t(u) \geq d_T(u, v).$$

The map is noncontracting. An upper Lipschitz bound on edges extends to all vertex pairs by the triangle inequality along their tree path.

Suppose a parent has $j \geq 1$ levels beneath it. Its edge to the first child has stretch 1. Before reaching the second child, the traversal explores the entire first-child branch and returns to the parent. The entering edge of each branch has length q^{j-1}; the total weight of one branch, including that edge, divided by q^{j-1} is

$$\sum_{i=0}^{j-1} (2/q)^i.$$

Consequently the stretch on the second-child edge is

$$L_j = 1 + 2 \sum_{i=0}^{j-1} (2/q)^i \leq L_h,$$

because $2/q > 1$. Summing the geometric series gives

$$L_h = 1 + \frac{2}{2-q} \left(\frac{N}{\gamma} - q \right) \leq 1 + 8 \left(\frac{N}{\gamma} - q \right) \leq \frac{8N}{\gamma}.$$

Here $N/\gamma - q > 0$, $2/(2-q) \leq 8$, and $q \geq 3/2$. The first-arrival map has distortion at most L_h, proving the upper bound. □

A.5 Metric-preserving graph witnesses

Let G have positive edge weights and a connected underlying undirected graph. We use the shortest-path metric d_G of that underlying graph, including when the edges of G carry a DAG orientation.

Proposition A.5 (Transfer through a metric witness)**.** *Let $T_\star$ be a positive weighted tree. Suppose an injective map $\iota : V(T_\star) \to V(G)$ satisfies*

$$a\, d_{T_\star}(u, v) \leq d_G(\iota(u), \iota(v)) \leq a\tau\, d_{T_\star}(u, v) \qquad (u, v \in V(T_\star)) \tag{20}$$

for some $a > 0$ and $\tau \geq 1$. If $S \subseteq V(T_\star)$ has $N \geq 2$ elements, pairwise separation at least s, and enclosing radius R_S as in (13)*, then every embedding of G into $\mathbb{R}^d$ with distortion C satisfies*

$$C \geq \frac{1}{\tau} \frac{s}{2R_S} \left(N^{1/d} - 1 \right). \tag{21}$$

Proof. Normalize the graph embedding f to be 1-Lipschitz. The assumed distance inequalities imply

$$\frac{a}{C} d_{T_\star}(u, v) \leq \|f(\iota(u)) - f(\iota(v))\|_2 \leq a\tau d_{T_\star}(u, v).$$

Consequently $f \circ \iota$ is an embedding of $T_\star$ with distortion at most $C\tau$. Apply Theorem A.1 to this composition and divide by τ. □

For a single packing witness, it suffices to check fewer distances. Let z be its center and suppose

$$d_G(\iota(u), \iota(v)) \geq as \quad (u \neq v \in S), \qquad d_G(\iota(z), \iota(u)) \leq a\tau R \quad (u \in S).$$

These conditions give a graph packing witness with separation as and enclosing radius $a\tau R$, so the same volume proof gives $C \geq s(N^{1/d} - 1)/(2\tau R)$. Thus the transfer can be certified directly on selected graph distances.

A.6 Necessary dimension and constructive sufficiency

Let $C_0 \geq 1$ be the distortion target, and let $\mathcal{S}$ be any nonempty collection of subsets of V with at least two elements. Define

$$d_{\mathrm{LB}}(T, C_0; \mathcal{S}) = \max\left\{1, \left\lceil \max_{S \in \mathcal{S}} \frac{\log |S|}{\log(1 + 2R_S C_0 / s(S))} \right\rceil\right\}. \tag{22}$$

If a distortion-C_0 embedding exists in $\mathbb{R}^d$, then $d \geq d_{\mathrm{LB}}(T, C_0; \mathcal{S})$. Indeed, Theorem A.1 gives, separately for each S,

$$\log |S| \leq d \log(1 + 2R_S C_0 / s(S)).$$

The denominator is positive; dividing, maximizing, and using integer dimension proves the claim. An upper bound on R_S or a lower bound on $s(S)$ gives a valid, possibly weaker, certificate. The maximum can be taken over all eligible subsets for the strongest bound of this form. For an observed domain O, use $S \subseteq O$ and $R_{S;O}$ instead.

The balanced midpoint witness yields the necessary real-valued threshold

$$d \geq \frac{\lfloor h/2 \rfloor \log b}{\log(1 + 2C_0\gamma)}, \tag{23}$$

while the full collection in (17) gives

$$d \geq \max_{1 \leq k \leq h} \frac{k \log b}{\log\left(1 + \dfrac{h\gamma C_0}{h - k + 1}\right)}. \tag{24}$$

A sufficient condition follows from an explicit Euclidean construction.

Proposition A.6 (Edge-coordinate embedding)**.** *Let $m = |E|$, and define $F : V \to \mathbb{R}^m$ by*

$$F(v)_e = w_e \, \mathbf{1}\{e \in \mathrm{path}(\mathrm{root}, v)\}.$$

Its distortion is exactly

$$K_T = \max_{u \neq v} \frac{\sum_{e \in \mathrm{path}(u,v)} w_e}{\left(\sum_{e \in \mathrm{path}(u,v)} w_e^2\right)^{1/2}} \leq \sqrt{q_T} \leq \sqrt{\min\{m, 2h\}}, \tag{25}$$

where q_T is the maximum number of edges in a vertex-to-vertex path. In particular, $d \geq m$ and $C_0 \geq K_T$ guarantee a distortion-C_0 embedding into $\mathbb{R}^d$.

Proof. The common initial root path cancels in $F(u) - F(v)$. Each remaining coordinate corresponds to one edge of the u–v path and has magnitude w_e. Therefore

$$\|F(u) - F(v)\|_2^2 = \sum_{e \in \mathrm{path}(u,v)} w_e^2.$$

This is positive for distinct vertices. The inequality $\sqrt{\sum w_e^2} \leq \sum w_e$ proves that F is 1-Lipschitz. Equality holds on each edge, so its Lipschitz constant is exactly 1 and its distortion is the displayed K_T. By Cauchy–Schwarz,

$$\sum_{e \in \mathrm{path}(u,v)} w_e \leq \sqrt{|\,\mathrm{path}(u,v)|} \left(\sum_{e \in \mathrm{path}(u,v)} w_e^2\right)^{1/2}.$$

Every path has at most $q_T \leq \min\{m, 2h\}$ edges. Finally, padding with zero coordinates gives an embedding in every dimension $d \geq m$. □

B Hyperbolic construction and proof

We give the construction and proof of Theorem 4.1, including the effects of edge weights, curvature, and an absolute separation requirement. Let o be the root, let $w_{\min} > 0$ be the minimum edge length, and write

$$t_v = d_T(o, v), \qquad H = \max_{v \in V(T)} t_v, \qquad B \geq \max\{2, \max_{v \in V(T)} \deg^+(v)\}.$$

Thus H is weighted height, while B bounds local fanout.

B.1 Nested angular intervals

First work in $\mathbb{H}^2_{-1}$ with polar metric

$$ds^2 = dr^2 + \sinh^2(r)\, d\theta^2. \tag{26}$$

Choose $\tau > 0$ with $\tau w_{\min} \geq \log(4B)$ and define

$$r_v = \tau t_v, \qquad A_v = e^{-r_v}.$$

Each vertex receives an angular interval I_v of width A_v. Assign the root $I_o = [-1/2, 1/2]$, in radians, and set $\theta_o = 0$. Suppose

$$I_v = [\theta_v - A_v/2, \theta_v + A_v/2]$$

has been assigned and v has children $v_1, \ldots, v_m$, in any fixed order. Put

$$\theta_{v_i} = \theta_v + A_v \left(\frac{i - 1/2}{m} - \frac{1}{2} \right), \qquad I_{v_i} = [\theta_{v_i} - A_{v_i}/2, \theta_{v_i} + A_{v_i}/2], \tag{27}$$

where $A_{v_i} = A_v e^{-\tau w(v, v_i)}$. Place $f(v)$ at polar coordinates (r_v, θ_v). The angular coordinate of o is immaterial because $r_o = 0$. These formulas specify the vertex embedding completely; a tree edge connects its two embedded endpoints by their hyperbolic geodesic.

Each child interval has width at most $A_v/(4B)$. The centers in (27) are the midpoints of m equal cells of I_v. Their distance to the boundary of the relevant cell is $A_v/(2m) \geq A_v/(2B)$, greater than the child interval's half-width $A_v/(8B)$. Hence $I_{v_i} \subseteq I_v$. Two distinct child centers are separated by at least $A_v/m \geq A_v/B$, so their intervals have a gap of at least

$$\frac{A_v}{B} - \frac{A_v}{4B} = \frac{3A_v}{4B}. \tag{28}$$

The intervals of descendants remain nested inside those of their ancestors.

Consequently, if $z = \mathrm{LCA}(u, v)$ and $\delta = |\theta_u - \theta_v|$, then

$$0 \leq \delta \leq e^{-r_z} \leq 1. \tag{29}$$

If neither vertex is an ancestor of the other, u and v belong to different child subtrees of z, and

$$\delta \geq \frac{3e^{-r_z}}{4B}. \tag{30}$$

These estimates apply to balanced and unbalanced trees and to arbitrary positive edge weights.

B.2 Distance estimates

Lemma B.1 (A polar distance inequality)**.** *Two points of $\mathbb{H}^2_{-1}$ with radii $r, s \geq 0$ and angular separation $0 < \delta \leq \pi$ satisfy*

$$d_{-1}(r, s, \delta) \geq r + s + 2 \log \sin(\delta/2). \tag{31}$$

Proof. Write $d = d_{-1}(r, s, \delta)$, $q = \sin(\delta/2)$, $U = (r + s)/2$, and $V = (r - s)/2$. The hyperbolic law of cosines [9, 10] is

$$\cosh d = q^2 \cosh(2U) + (1 - q^2) \cosh(2V).$$

The half-angle identities therefore give

$$\sinh^2(d/2) = q^2 \sinh^2 U + (1-q^2)\sinh^2 V \geq q^2 \sinh^2 U,$$
$$\cosh^2(d/2) = q^2 \cosh^2 U + (1-q^2)\cosh^2 V \geq q^2 \cosh^2 U.$$

Taking nonnegative square roots and adding yields $e^{d/2} \geq qe^U$. Taking logarithms proves the assertion. □

Lemma B.2 (Additive distortion bound)**.** *The interval construction with $\tau w_{\min} \geq \log(4B)$ satisfies*

$$\tau d_T(u,v) - 2\log(4B) \leq d_{-1}(f(u), f(v)) \leq \tau d_T(u,v) + \frac{1}{2} \quad \textit{for all } u, v \in V(T). \tag{32}$$

Proof. Let $z = \mathrm{LCA}(u,v)$. For the upper bound, follow the root ray of $f(u)$ inward to radius r_z, traverse the circle of radius r_z through the angle δ, and follow the other root ray outward to $f(v)$. By (26) and (29), this path has length

$$\begin{aligned} r_u + r_v - 2r_z + \sinh(r_z)\delta &\leq \tau d_T(u,v) + \sinh(r_z)e^{-r_z} \\ &\leq \tau d_T(u,v) + \frac{1}{2}. \end{aligned}$$

The distance between its endpoints is no larger than its length.

For the lower bound, if one vertex is an ancestor of the other, the reverse triangle inequality with respect to the embedded root gives

$$d_{-1}(f(u), f(v)) \geq |r_u - r_v| = \tau d_T(u,v).$$

Otherwise, combine Lemma B.1 with (30). Since $\delta \leq 1$, the inequality $\sin(\delta/2) \geq \delta/3$ applies; it follows, for example, from $\sin x \geq x - x^3/6$ on $[0, 1/2]$. Thus

$$\begin{aligned} d_{-1}(f(u), f(v)) &\geq r_u + r_v + 2\log\sin(\delta/2) \\ &\geq r_u + r_v + 2\log\frac{e^{-r_z}}{4B} \\ &= \tau d_T(u,v) - 2\log(4B). \end{aligned}$$

The case $u = v$ is immediate. □

Proof of Theorem 4.1. For distinct vertices, $d_T(u,v) \geq w_{\min}$. If

$$\tau w_{\min} \geq \frac{2\log(4B)}{\eta}, \qquad 0 < \eta < 1,$$

both additive errors in (32) are at most $\eta\tau d_T(u,v)$, because $1/2 \leq 2\log(4B)$. Therefore

$$(1-\eta)\tau d_T(u,v) \leq d_{-1}(f(u), f(v)) \leq (1+\eta)\tau d_T(u,v).$$

The same inequality holds for identical vertices. The hyperbolic plane of curvature $-\kappa$ is obtained by multiplying all distances in the curvature--1 model by $1/\sqrt{\kappa}$. Accordingly,

$$\frac{(1-\eta)\tau}{\sqrt{\kappa}} d_T(u,v) \leq d_{-\kappa}(f(u), f(v)) \leq \frac{(1+\eta)\tau}{\sqrt{\kappa}} d_T(u,v).$$

The ratio of the upper and lower multiplicative constants bounds $\mathrm{dist}(f)$ by $(1+\eta)/(1-\eta)$. The distance of $f(v)$ from the root is exactly $r_v/\sqrt{\kappa}$, so the largest such distance is $\tau H/\sqrt{\kappa}$. □

B.3 A joint radius and separation certificate

For a prescribed distortion $C_0 > 1$, set

$$\eta = \frac{C_0 - 1}{C_0 + 1}, \qquad \tau \geq \frac{2\log(4B)}{\eta w_{\min}}.$$

Theorem 4.1 then yields $\mathrm{dist}(f) \leq C_0$. We distinguish the scale-invariant distortion target from an absolute separation requirement.

Corollary B.3 (Sufficient budget with an absolute margin)**.** *Let $s_0 > 0$ be the required minimum hyperbolic distance between every pair of distinct vertices, and let $R > 0$ be a radius budget. The construction meets these requirements and has distortion at most C_0 whenever*

$$\frac{\tau H}{R} \le \sqrt{\kappa} \le \frac{(1-\eta)\tau w_{\min}}{s_0}. \tag{33}$$

The interval in (33) *is nonempty if and only if*

$$R \ge R_{\mathrm{cert}} := \frac{H s_0}{(1-\eta) w_{\min}}. \tag{34}$$

Proof. The exact radius formula gives the left inequality in (33). Since distinct vertices have tree distance at least $w_{\min}$, the theorem guarantees separation at least $(1-\eta)\tau w_{\min}/\sqrt{\kappa}$, yielding the right inequality. The interval is nonempty exactly when its lower endpoint does not exceed its upper endpoint. Cancelling $\tau > 0$ gives (34). □

Here R_{cert} is a sufficient budget for the stated construction and separation guarantee. For fixed κ, the certificate is checked directly using (33). If curvature is adjustable and only scale-invariant distortion and physical radius are constrained, any positive radius can be met by increasing κ. An absolute margin such as s_0 prevents this rescaling from making all embedded distances arbitrarily small.

B.4 Curvature and numerical representation

The largest dimensionless radius is $M = \tau H$. In the normalized unit Poincaré disk, a point at physical radius $r_v/\sqrt{\kappa}$ has Euclidean norm

$$\rho_v = \tanh(r_v/2), \qquad \rho_{\max} = \tanh(M/2).$$

These normalized coordinates and the scale-invariant distortion do not change when κ changes. Thus a smaller physical radius obtained by curvature rescaling does not reduce proximity to the disk boundary. For example,

$$1 - \rho_{\max} = \frac{2}{e^M + 1},$$

so direct fixed-point representation near the boundary requires precision that grows with M. This dependence must be included when comparing storage and arithmetic costs.

The angular construction also requires care at large depth. Define the normalized offset of child v_i of v as

$$o_{v,i} = \frac{i - 1/2}{m_v} - \frac{1}{2}.$$

For a descendant u of z, its angular displacement is

$$\theta_u - \theta_z = e^{-r_z} \sum_{(v,v_i)\in P(z,u)} e^{-(r_v - r_z)} o_{v,i}. \tag{35}$$

For two vertices in different child subtrees of their least common ancestor, the difference of the normalized sums in (35) is at least $3/(4B)$ in magnitude. Computing this difference relative to the least common ancestor and retaining the outer factor logarithmically avoids subtracting nearly identical global angular coordinates. For an ancestor pair, the first nonzero offset can be used to rescale the sum; when all offsets vanish, the distance equals the radial difference.

The identity

$$\cosh d_{-1} = \cosh(r_u - r_v) + 2\sinh(r_u)\sinh(r_v)\sin^2(\delta/2)$$

allows distance evaluation through logarithms of nonnegative terms and stable inverse hyperbolic functions. The validation experiments evaluate distances from this tree-relative representation. Storage and query costs include the chosen precision, retained tree information, and distance-evaluation procedure.

C Random edge weights

C.1 Setup and packing

Let $T = (V, E)$ be a finite rooted tree of combinatorial height h. Its edge weights satisfy

$$0 < \varepsilon < 1, \qquad W_e \in [1-\varepsilon, 1+\varepsilon], \qquad \mathbb{E}W_e = 1. \tag{36}$$

Write d_W for the resulting tree metric. Fix $1 \le r \le h$ and a set $\mathcal{S}_r$ of N leaves such that, for distinct $u, v \in \mathcal{S}_r$, the u–v path contains both terminal r-edge paths A_u, A_v. Equivalently, both leaves have at least r edges below their lowest common ancestor. This condition holds for equal-depth leaves with edge-disjoint terminal paths. The family is chosen from the topology before observing the weights.

For $0 < \delta \le \varepsilon$, define

$$X_u = \sum_{e \in A_u} W_e, \qquad Z_u = \mathbf{1}\{X_u \ge (1-\delta)r\}, \qquad Y = \sum_{u \in \mathcal{S}_r} Z_u, \qquad a_r = \frac{(1-\delta)r}{(1+\varepsilon)h}.$$

Lemma C.1 (Packing good leaves)**.** *On any event where $Y \ge m > 0$, every embedding $\psi : (V, d_W) \to \mathbb{R}^d$ with $\operatorname{dist}(\psi) \le C$ satisfies*

$$C \ge a_r\big(m^{1/d} - 1\big). \tag{37}$$

Proof. Rescale ψ to have Lipschitz constant one. Its co-Lipschitz constant is then at least $1/C$, and every image lies within radius $R = (1+\varepsilon)h$ of the root image. For good leaves, the critical-family condition gives $d_W(u, v) \ge X_u + X_v \ge 2(1-\delta)r$. Their images are therefore separated by $s = 2(1-\delta)r/C$. Disjoint open balls of radius $s/2$ around these images lie within a ball of radius $R + s/2$. Comparing volumes yields $Y \le (1 + 2R/s)^d = (1 + C/a_r)^d$. Since $Y \ge m$, rearrangement proves the claim. This event depends only on the weights, so its conclusion holds for all embeddings simultaneously. □

C.2 Independent weights

Proof of Theorem 5.1. Hoeffding's inequality [11] applied to the r independent centered variables $W_e - 1 \in [-\varepsilon, \varepsilon]$ gives

$$\mathbb{P}\{X_u < (1-\delta)r\} \le q := \exp\left(-\frac{\delta^2 r}{2\varepsilon^2}\right). \tag{38}$$

Let $p = 1 - q$. The terminal paths are edge-disjoint, so their indicators Z_u are independent, with $\mu := \mathbb{E}Y \ge pN$. Identical distributions are unnecessary. For $t > 0$,

$$\mathbb{E}e^{-tY} = \prod_u \big[1 - \mathbb{E}Z_u + \mathbb{E}Z_u e^{-t}\big] \le \exp\big(\mu(e^{-t} - 1)\big).$$

Exponential Markov inequality with $t = -\log(1-\theta)$, where $0 < \theta < 1$, gives

$$\mathbb{P}\{Y \le (1-\theta)\mu\} \le \exp\{-\mu[\theta + (1-\theta)\log(1-\theta)]\} \le e^{-\theta^2\mu/2}.$$

The last inequality follows from $\theta + (1-\theta)\log(1-\theta) \ge \theta^2/2$. Since $\mu \ge pN$, the probability that $Y < (1-\theta)pN$ is at most $e^{-\theta^2 pN/2}$. Lemma C.1, with $m = (1-\theta)pN$, completes the proof. □

The choice $\delta = \varepsilon/2$ gives $q = e^{-r/8}$. Taking $\theta = 1/2$ yields the good-count threshold $pN/2$ and failure probability $e^{-pN/8}$.

C.3 Local dependence

A *dependency graph* G_E for the weights has vertex set E and the following property: whenever disjoint sets $I, J \subseteq E$ have no graph edge between them, the random vectors $(W_e)_{e\in I}$ and $(W_e)_{e\in J}$ are independent. In particular, the variables in each color class of a proper coloring are mutually independent. Let a be the number of colors in such a coloring. Define H_r on $\mathcal{S}_r$ by joining u, v whenever G_E has an edge between A_u and A_v, and let c be the number of colors in a proper coloring of H_r.

Proposition C.2 (Dependency-graph bound)**.** *Under* (36) *and the dependency-graph assumption, put*

$$q = \exp\!\left(-\frac{\delta^2 r}{2\varepsilon^2 a}\right), \qquad p = 1 - q, \qquad m = (1-\theta)pN, \tag{39}$$

where $0 < \theta < 1$. *With probability at least* $1 - \exp(-\theta^2 pN/(2c))$*, one has* $Y \geq m$ *and every distortion-*C *embedding satisfies* (37)*. If* G_E *has maximum degree* D*, one may take* $a = D + 1$ *and* $c = rD + 1$.

Proof. For a terminal path, partition its centered weights $V_e = W_e - 1$ into the a color classes I_j. Generalized Hölder's inequality, independence within a class, and Hoeffding's lemma give

$$\mathbb{E}\exp\!\left(-t\sum_{e\in A_u} V_e\right) \leq \prod_{j=1}^{a}\left[\mathbb{E}\exp\!\left(-at\sum_{e\in I_j} V_e\right)\right]^{1/a} \leq \exp(ar\varepsilon^2 t^2/2).$$

Exponential Markov inequality, minimized at $t = \delta/(a\varepsilon^2)$, gives the tail bound q in (39). Consequently $\mu = \mathbb{E}Y \geq pN$.

The graph H_r is a dependency graph for the indicators: collections with no connecting H_r edge depend on edge sets with no connecting G_E edge. Applying Hölder to its c color classes yields

$$\mathbb{E}e^{-tY} \leq \prod_{u}\left(\mathbb{E}e^{-ctZ_u}\right)^{1/c} \leq \exp\!\left[\frac{\mu}{c}(e^{-ct} - 1)\right].$$

Taking $ct = -\log(1-\theta)$ in the preceding Chernoff argument bounds $\mathbb{P}\{Y < (1-\theta)pN\}$ by $\exp(-\theta^2 pN/(2c))$. Apply Lemma C.1.

Finally, greedy coloring uses at most $D+1$ colors in G_E. Each terminal path contains r edges, each with at most D neighbors. Since terminal paths are edge-disjoint, the degree of H_r is at most rD; greedy coloring then uses at most $rD + 1$ colors. □

For dependence of range L, take G_E to join tree edges at line-graph distance at most L, and require independence of edge sets with no such connections. If the line graph has maximum degree Q, the explicit bound $D \leq \sum_{j=1}^{L} Q^j$ depends only on L and the branching bound. The resulting constants account for dependence both within a terminal path and between different paths.

C.4 Geometric mixing

We use the standard strong-mixing coefficient [12]. For sigma-algebras $\mathcal{A}, \mathcal{B}$, write

$$\alpha(\mathcal{A}, \mathcal{B}) = \sup_{A\in\mathcal{A},\, B\in\mathcal{B}} |\mathbb{P}(A\cap B) - \mathbb{P}(A)\mathbb{P}(B)|.$$

Assume that along every root-to-leaf path $(e_1, \ldots, e_k)$,

$$\alpha\big(\sigma(W_{e_1}, \ldots, W_{e_i}), \sigma(W_{e_{i+j}}, \ldots, W_{e_k})\big) \leq \rho^j \quad (i + j \leq k), \qquad 0 < \rho < 1. \tag{40}$$

Stationarity is not needed. Let ℓ_{uv} denote the minimum line-graph distance between A_u and A_v.

Proposition C.3 (Mixing bound)**.** *Under* (36) *and* (40)*, set*

$$v_r = 1 + \frac{8}{r}\sum_{j=1}^{r-1}(r-j)\rho^j \leq 1 + \frac{8\rho}{1-\rho}, \qquad q = \frac{\varepsilon^2 v_r}{\varepsilon^2 v_r + \delta^2 r}, \qquad p = 1 - q. \tag{41}$$

If, additionally,

$$\alpha\big(\sigma(W_e : e \in A_u), \sigma(W_e : e \in A_v)\big) \le \rho^{\ell_{uv}} \quad (u \neq v), \tag{42}$$

define $V = N/4 + 2\sum_{\{u,v\}\subset \mathcal{S}_r} \rho^{\ell_{uv}}$. *For* $0 < \theta < 1$, *with probability at least*

$$1 - \frac{V}{V + \theta^2 p^2 N^2}, \tag{43}$$

one has $Y \ge m = (1-\theta)pN$, *and every distortion-*C *embedding satisfies* (37).

Without (42), *one still has* $Y \ge (1-\theta)N$ *with probability at least* $1 - q/\theta$. *On that event,* (37) *holds with* $m = (1-\theta)N$.

Proof. For variables $U, V \in [-\varepsilon, \varepsilon]$ measurable with respect to sigma-algebras of mixing coefficient at most b,

$$|\operatorname{Cov}(U, V)| \le 4\varepsilon^2 b. \tag{44}$$

Indeed, the layer-cake representations of $U + \varepsilon, V + \varepsilon$ express their covariance as the integral over $[0, 2\varepsilon]^2$ of $\mathbb{P}(U + \varepsilon > s, V + \varepsilon > t) - \mathbb{P}(U + \varepsilon > s)\mathbb{P}(V + \varepsilon > t)$, whose absolute value is at most b. Thus (40), together with $\operatorname{Var}(W_e) \le \varepsilon^2$, gives

$$\operatorname{Var}(X_u) \le r\varepsilon^2 + 8\varepsilon^2 \sum_{j=1}^{r-1} (r-j)\rho^j = \varepsilon^2 r v_r.$$

For a centered variable F of variance σ^2, Cantelli's inequality is $\mathbb{P}(F \le -t) \le \sigma^2/(\sigma^2 + t^2)$. It follows by applying Markov's inequality to $(F - b)^2$ and choosing $b = \sigma^2/t$. Apply it with $F = X_u - r$ and $t = \delta r$ to obtain $\mathbb{P}(Z_u = 0) \le q$, hence $\mu = \mathbb{E}Y \ge pN$.

Under (42), the indicators satisfy $|\operatorname{Cov}(Z_u, Z_v)| \le \rho^{\ell_{uv}}$ directly from the definition of α, and $\operatorname{Var}(Z_u) \le 1/4$. Therefore $\operatorname{Var}(Y) \le V$. The event $Y < (1-\theta)pN$ implies $Y - \mu < -\theta pN$. A second application of Cantelli gives the failure bound in (43); packing completes the first conclusion. For the final conclusion, the number of bad leaves $B = N - Y$ has $\mathbb{E}B \le qN$. Markov's inequality gives $\mathbb{P}(B > \theta N) \le q/\theta$, without any assumption between different terminal paths. □

For an explicit summary of cross-branch dependence, let $K = \max_u \sum_{v \neq u} \rho^{\ell_{uv}}$. Then $V \le N(1/4 + K)$, so when K is bounded and p is bounded away from zero, (43) has failure probability $O(N^{-1})$. The distortion bound retains its $N^{1/d}$ dependence.

C.5 Simultaneous scale selection

Corollary C.4 (Selection over a finite family)**.** *Fix a finite collection* $\mathcal{J}$ *of candidate critical families, scales, and parameters before observing the weights. For candidate* j, *let* $m_j > 0$ *be a good-count threshold above,* β_j *its failure bound, and* $a_j = (1 - \delta_j)r_j/((1+\varepsilon)h)$. *With probability at least* $1 - \sum_{j\in\mathcal{J}} \beta_j$, *every distortion-*$C$ *embedding into* $\mathbb{R}^d$ *satisfies*

$$C \ge \max_{j\in\mathcal{J}} a_j (m_j^{1/d} - 1). \tag{45}$$

On the same event, any embedding with distortion at most C_0 *must have*

$$d \ge d_{\mathrm{LB}} := \max\left\{1, \left\lceil \max_{j\in\mathcal{J}:m_j>1} \frac{\log m_j}{\log(1 + C_0/a_j)} \right\rceil\right\}, \tag{46}$$

with the empty inner maximum defined as zero.

Proof. The union bound makes all events $Y_j \ge m_j$ simultaneous with the stated probability. Apply Lemma C.1 to each candidate and take their maximum. For $m_j > 1$, solving $C_0 \ge a_j(m_j^{1/d} - 1)$ for d, then taking the maximum and rounding up, proves (46). □

One may optimize over this fixed collection after observing the weights while retaining the simultaneous guarantee. A prescribed failure budget β is met whenever $\sum_j \beta_j \le \beta$. The resulting d_{LB} is a necessary dimension: a budget below it is ruled out at the stated confidence.

D Experimental details and reproducibility

D.1 Tree families and sampling

The balanced family is a complete binary tree of height six. The spine-and-bush family starts with a path of six edges. We attach one complete binary subtree of height three to each of its seven vertices and attach a second such subtree to the final spine vertex. Each attached subtree has 15 new vertices, including its root; its entering edge connects it to the spine. Both families have 127 vertices and 64 leaves; their maximum combinatorial depths are six and ten, respectively. Each edge initially receives an independent $\mathrm{Unif}[1, 2]$ weight. The clustered-weight family uses the balanced topology and multiplies every edge in one root branch, including its entering edge, by four; the other branch is multiplied by one quarter. There are 30 independently weighted instances per family. The weight generator uses seeds $71000 + s$, $s = 0, \ldots, 29$.

D.2 Euclidean optimization

We fit metric MDS to the complete tree distance matrix, including every vertex. The objective is the unweighted raw stress $\sum_{u<v}(\|x_u - x_v\| - d_T(u, v))^2$. We run complete-graph SMACOF from three starts: classical MDS and two independent Gaussian initializations. Each start runs for at most 1,000 iterations, stopping earlier when the stress reduction is at most $10^{-7} \max\{1, \text{previous stress}\}$. We retain the start with the smallest final stress. The random initialization seed is $83000 + 100s + d$. The objective and final distortion are separate quantities: selecting a smaller stress need not select a smaller worst-case distortion. Final distances use direct coordinate differences to preserve very small separations. In the primary run, 269 of 270 two-dimensional starts met the stopping criterion; all five- and ten-dimensional starts reached the iteration cap. A 5,000-iteration sensitivity check on the first instance of each family changes the resulting distortion by less than 4%; Table 2 reports the results.

D.3 Distortion and packing bounds

For each fitted embedding we calculate $a = \min_{u<v} d_Y(f(u), f(v))/d_T(u, v)$ and $b = \max_{u<v} d_Y(f(u), f(v))/d_T(u, v)$, then report b/a. Thus the statistic is unchanged by a common rescaling of the embedding. Every primary tree contributes all $127 \cdot 126/2 = 8001$ pairs. Reported medians and interquartile ranges describe variation across the 30 independently weighted instances.

For each combinatorial depth $k \geq 1$, its nodes form an antichain. Below each node v we choose a leaf of maximum weighted root distance. Its terminal path starts at $\mathrm{parent}(v)$ and therefore includes the edge entering v. Let ℓ_v be that path's weight. We sort these lengths and, for each $m \geq 2$, keep the m largest. The selected leaf pairs have distance at least $2\ell_{(m)}$, so the computed lower bound is

$$\max\left\{1,\ \max_{k,m} \frac{\ell_{(m)}}{H}(m^{1/d} - 1)\right\}.$$

D.4 Hyperbolic distances and numerical checks

We use the construction in Theorem 4.1 with $\eta = 1/3$ and curvature -1. For all 8,001 pairs of each tree, the implementation checks both additive bounds and the resulting distortion bound of two. Global angles can be almost equal when nodes share a deep ancestor. We instead form their angular difference in the coordinate interval of their lowest common ancestor and evaluate

$$\cosh d_H = \cosh(r_u - r_v) + 2 \sinh r_u \sinh r_v \sin^2((\theta_u - \theta_v)/2)$$

in the logarithmic domain. This is an evaluation of the specified embedding using its tree representation; it does not assume that two ordinary floating-point coordinates preserve every angular separation. Our implementation stores the tree paths, lowest-common-ancestor matrix, and relative angles. A direct implementation uses $O(n^2)$ auxiliary space and up to $O(n^2 h)$ preprocessing; exhaustive distance evaluation then takes $O(n^2)$ operations. These numerical evaluation costs are separate from the real-arithmetic embedding's two-dimensional target space.

An independent high-precision check evaluates global polar coordinates and the hyperbolic cosine law on three small weighted trees, including unary paths. The logarithmic implementation agrees

with a maximum relative error below 7.2×10^{-16} in those checks. The same validation script compares tree distances with an independent all-pairs shortest-path computation, checks the packing separation premise, and recovers an exact Euclidean test metric with normalized raw stress below 3×10^{-32}.

D.5 Depth, weights, and iteration budget

The depth sweep uses complete unit-weight binary trees with heights three through seven and 15, 31, 63, 127, and 255 vertices. The weight sweep uses one height-five binary tree with weights drawn independently from $\mathrm{Unif}[1, 2]$ using seed 95000. We hold those weights fixed and multiply every edge of one root branch by $1, 2, 4, 8$, or 16. All other settings match the primary experiment.

Table 2: Iteration-budget sensitivity on the first weighted instance of each family. Each entry gives all-pairs distortion after at most 1,000 / 5,000 SMACOF iterations per start; all other settings are unchanged. Selection minimizes raw stress.

Family	$d = 2$	$d = 5$	$d = 10$
Balanced	38.78 / 38.78	6.08 / 6.10	2.99 / 2.92
Spine-and-bush	41.95 / 41.95	5.56 / 5.78	3.07 / 3.10
Clustered weights	43.20 / 43.20	12.17 / 12.09	4.38 / 4.37

D.6 Parent and sibling retrieval

For each of the 64 leaves, we rank all 63 internal vertices by embedded distance and record the rank of its true parent. Hits@1 averages the indicator of rank one; MRR averages the reciprocal rank. Equal distances are ordered by vertex index. We also report whether the nearest other leaf has the same parent. These retrieval statistics are computed from the same complete embeddings used for distortion. Table 3 reports the mean of each score over the 30 instances.

Table 3: Parent retrieval from the same complete embeddings. Entries are means over 30 independently weighted trees. Each leaf ranks all 63 internal vertices; sibling accuracy measures whether its nearest other leaf has the same parent.

Family	Embedding	Hits@1	MRR	Sibling accuracy
Balanced	MDS, $d = 2$	0.323	0.578	0.637
Balanced	MDS, $d = 5$	0.994	0.997	0.998
Balanced	MDS, $d = 10$	1.000	1.000	1.000
Balanced	Constructive $\mathbb{H}^2$	1.000	1.000	1.000
Spine-and-bush	MDS, $d = 2$	0.329	0.587	0.644
Spine-and-bush	MDS, $d = 5$	0.965	0.982	1.000
Spine-and-bush	MDS, $d = 10$	1.000	1.000	1.000
Spine-and-bush	Constructive $\mathbb{H}^2$	1.000	1.000	1.000
Clustered weights	MDS, $d = 2$	0.515	0.741	0.802
Clustered weights	MDS, $d = 5$	0.776	0.887	0.936
Clustered weights	MDS, $d = 10$	0.998	0.999	1.000
Clustered weights	Constructive $\mathbb{H}^2$	1.000	1.000	1.000

D.7 Independent edge noise

The chosen terminal-path family is fixed before any edge weights are sampled. Directly sampling its disjoint paths has the same joint law as sampling the full binary tree and retaining those path sums. With $r = h/2$, $N = 2^{h-r}$, $\delta = \varepsilon/2$, and count slack $1/2$, let $p_0 = 1 - e^{-r/8}$. We measure the event $Y \geq \lceil Np_0/2 \rceil$, whose probability is at least $1 - e^{-Np_0/8}$. The per-path threshold $(1 - \varepsilon/2)r$ exceeds the minimum possible path weight $(1 - \varepsilon)r$. Exact uniform-model probabilities use the Irwin–Hall distribution for a path sum and the binomial distribution for the independent count. We keep these exact probabilities separate from the distribution-free lower bounds. Every setting has 10,000 independent trials. Zero observed count-event failures gives the two-sided 95%

Wilson interval $[0.999616, 1]$. A master seed of 12920261021 creates separate PCG64 streams for all nine settings; the stream seeds and count summaries are recorded in the accompanying results.

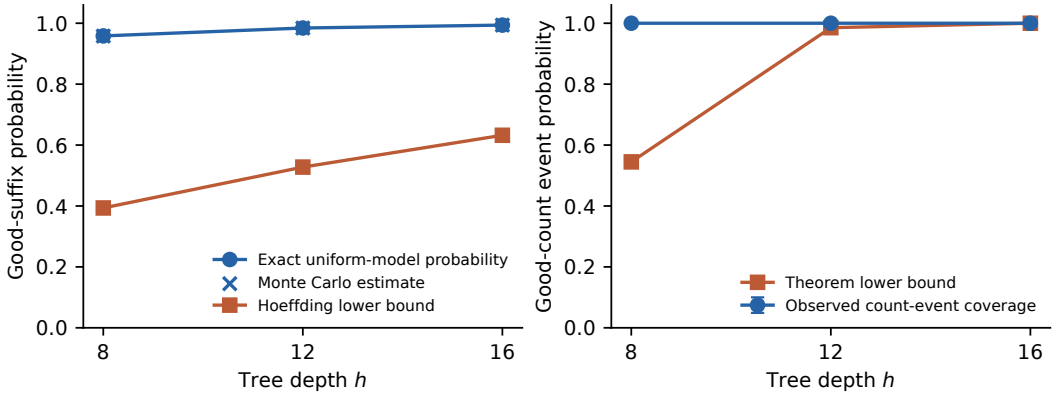


Figure 3: Independent edge noise on binary trees, with $r = h/2$, $\varepsilon = 0.3$, and $\delta = \varepsilon/2$. Each depth uses 10,000 trials. Left: the empirical probability that a terminal path is good agrees with the exact uniform-model calculation and exceeds its Hoeffding lower bound. Right: observed coverage of the good-count event exceeds the theorem's guaranteed probability; error bars are 95% Wilson intervals across independent trials.

D.8 Software and reproducibility

The primary experiment uses Python 3.13.5, NumPy 2.3.1, and Matplotlib; the noise illustration also uses SciPy. The source archive contains the complete experiment programs, per-instance CSV results, solver diagnostics, software metadata, and independent numerical validation. No GPU is required.